

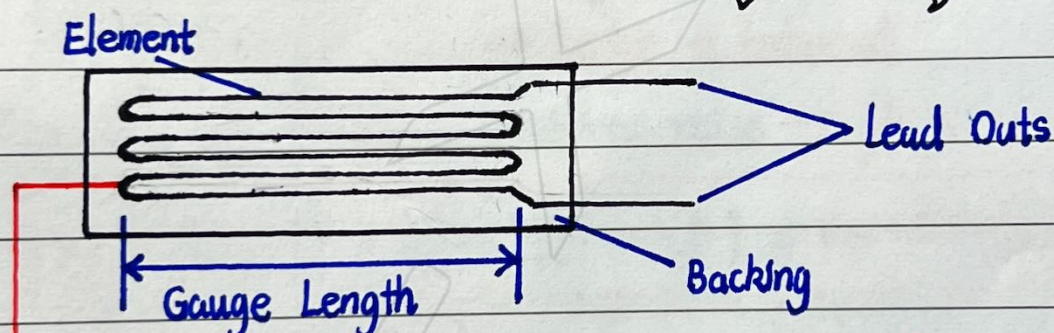
11/27/2022 Strain Measurement Pre-Lab

Resistance strain gauges

- $E_{\text{surface}} = E_{\text{strain gauge conductor}}$
- R of an electrical conductor varies if gauge deformed
- wire / acoustic 声音的 / capacitive 电容的 / inductive 感应的 / mechanical / optical 光学的 / piezoresistive 压阻的 / semi-conductive strain gauge

Wire strain gauge

- low cost • established (most commonly used) product



serpentine 弯曲的 wire mainly run in measured direction

→ maximise length of conductor

→ maximise ΔR due to strain ϵ in that direction

* Approximate gauge as a conductive wire of constant cross-section A

$$R = \rho \frac{l}{A}$$

ρ ———— resistivity of the material

l ———— wire length

A ———— cross-sectional area

If strain is not too large

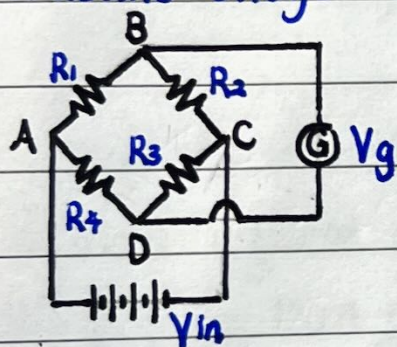
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If strain is not too large $\Delta R \propto \epsilon$

$$\frac{\Delta R^*}{R} = S \epsilon \quad \text{gauge factor} \leftarrow \approx 2.1 \text{ in this lab}$$

* $\Delta R \sim 0.1\%$ $\leftarrow$ difficult to measure $\leftarrow$ Wheatstone bridge circuit

Wheatstone bridge (electric circuit)



$$\Delta R = \Delta R_{\text{thermal}} + \Delta R_{\text{actual}}$$

$$R_1 = R_2 = R_3 = R_4 = R$$

$\rightarrow$ output voltage, $V_g = 0$

$\rightarrow$ bridge is balanced

After balanced $\rightarrow$ change resistors value $\rightarrow$ if Δ are relatively small

$$V_g = \frac{V_{in}}{4} \left[\frac{\Delta R_1}{R_1} - \frac{\Delta R_2}{R_2} + \frac{\Delta R_3}{R_3} - \frac{\Delta R_4}{R_4} \right] = V_{in} \frac{\Delta R_1 - \Delta R_2 + \Delta R_3 - \Delta R_4}{4R}$$

Replaced 1 'quarter-bridge' / 2 'half-bridge' / 4 'full-bridge' resistors

* if connect a strain gauge ($120/350 \Omega$) to first arm of bridge AB

$\rightarrow$ deformed $\rightarrow \Delta R_1$ while $\Delta R_2 = \Delta R_3 = \Delta R_4 = 0$

$$\rightarrow V_g = V_{in} \frac{\Delta R_1}{4R}$$

$$\rightarrow \epsilon = \frac{\frac{\Delta R}{R}}{S} = \frac{4V_g}{V_{in} S}$$

* V_g is quite small $\rightarrow$ connect the voltmeter to voltage amplifier

$$\rightarrow \bar{V}_g = \text{GAIN} \cdot V_g$$

$$\rightarrow \epsilon = \frac{4V_g}{V_{in} S} = \frac{4\bar{V}_g}{V_{in} S \cdot \text{GAIN}}$$

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$V_{in} \frac{\Delta R_1 - \Delta R_2 + \Delta R_3 - \Delta R_4}{4R}$ → if strap R_1 by gauge

$$\Delta R_1 = \Delta R_{thermal} + \Delta R_{actual}$$

→ strap R_2 by dummy gauge 补偿应变片

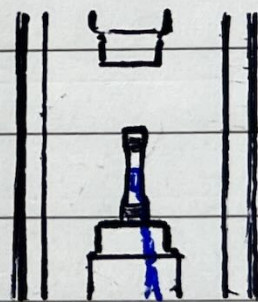
$$\Delta R_2 = \Delta R_{thermal}$$

$$\rightarrow V_g = V_{in} \frac{\Delta R_1 - \Delta R_2}{4R} = V_{in} \frac{\Delta R_{thermal} + \Delta R_{actual} - \Delta R_{thermal}}{4R} = V_{in} \frac{\Delta R_{actual}}{4R}$$

eliminate thermal effects ← thermal compensation 抵消

Laboratory Experiment

- II round drybone specimens
- strain gauge 120Ω in axial direction
- Place specimens in machine
- ↑ load 500N in steps up to 3.5kN (maximum)
- load > 3.5kN → break or yield specimen
→ invalid measure of E
- Wire connected to a simple Wheatstone bridge
- quarter-bridge configuration
- voltmeter - voltage amplifier record value



Procedure

- Measure D → calculate A

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- Record **S** and **GAIN**
- * Do not induced load when fasten the specimen's ends
- connect gauge wires to the Wheatstone bridge
- switch off the bridge $\rightarrow$ Record **V_{in}** by DC voltage voltmeter
- switch on the bridge $\rightarrow$ Read **$\bar{V}_g$**
- turn the dial of the potentiometer till **$\bar{V}_g \neq 0$** $\rightarrow$ balanced
- very slowly operate wheel of tensile machine
 $\rightarrow F = 0.5 \text{ kN} \rightarrow$ Record **$\bar{V}_g$** $\rightarrow$ calculate **tensile strain ϵ**
- Repeat till **$F = 3.5 \text{ kN} \sim 3 \text{ kN}$**
- Unload the specimen slowly until load drop to 0
- Remove specimen
- Repeat with next specimen
- Graph 1 **F vs $\bar{V}_g$**
- Graph 2 **σ vs ϵ**

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Gauge factors S	GAIN (V)	V_{in} (V)	Specimen	1st D (mm)	2nd D (mm)	3rd D (mm)	Specimen's diameter D (mm)	Number of attempt	State	Applied force F (kN)	GAIN* V_g (V)
			Aluminum					1	Load		
								2			
								3			
								4			
								5			
								6			
								7			
								8			
								9	Unload		
								10			
								11			
								12			
								13			
								14			
								15			
			Copper					1	Load		
								2			
								3			
								4			
								5			
								6			
								7			
								8			
								9	Unload		
								10			
								11			
								12			
								13			
								14			
								15			

* Copper do not
return to
zero even
after re-doing
the experiment

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Gauge factors S	GAIN (V)	V_{in} (V)	Specimen	1st D (mm)	2nd D (mm)	3rd D (mm)	Specimen's diameter D (mm)	Number of attempt	State	Applied force F (kN)	GAIN* V_g (V)		
			Brass					1	Load				
								2					
								3					
								4					
								5					
								6					
								7					
								8					
								9	Unload				
								10					
								11					
								12					
								13					
								14					
								15					
			Iron				1	Load					
							2						
							3						
							4						
							5						
							6						
							7						
							8						
							9	Unload					
							10						
							11						
							12						
							13						
							14						
							15						

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Gauge factors

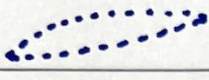
Specimen	Specimen's diameter D (mm)	Number of attempt	State	Applied force F (N)	V_g (V)	Stress σ (N/m ²)	Strain ϵ
Aluminum		1	Load		$V_g = GAIN * V_g / GAIN$	$\sigma = F / A$	$\epsilon = 4V_g / V_m * S$
		2					
		3					
		4					
		5					
		6					
		7					
		8					
		9	Unload				
		10					
		11					
		12					
		13					
		14					
		15					
Copper		1	Load				
		2					
		3					
		4					
		5					
		6					
		7					
		8					
		9	Unload				
		10					
		11					
		12					
		13					
		14					
		15					

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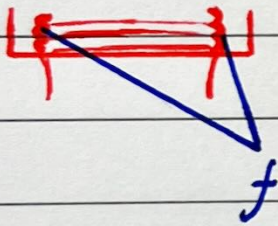
Gauge factors

Specimen	Speicimen's diameter D (mm)	Number of attempt	State	Applied force F (N)	$V_g(V)$ $V_g = GAIN * V_g / GAIN$	Stress σ (N/m ²) $\sigma = F/A$	Strain ϵ $\epsilon = 4V_g / V_{in} * S$
Brass		1	Load				
		2					
		3					
		4					
		5					
		6					
		7					
		8					
		9	Unload				
		10					
		11					
		12					
		13					
		14					
		15					
Iron		1	Load				
		2					
		3					
		4					
		5					
		6					
		7					
		8					
		9	Unload				
		10					
		11					
		12					
		13					
		14					
		15					

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Shape 
occurs because

- human error
- systematic error
- energy dissipation
due to friction



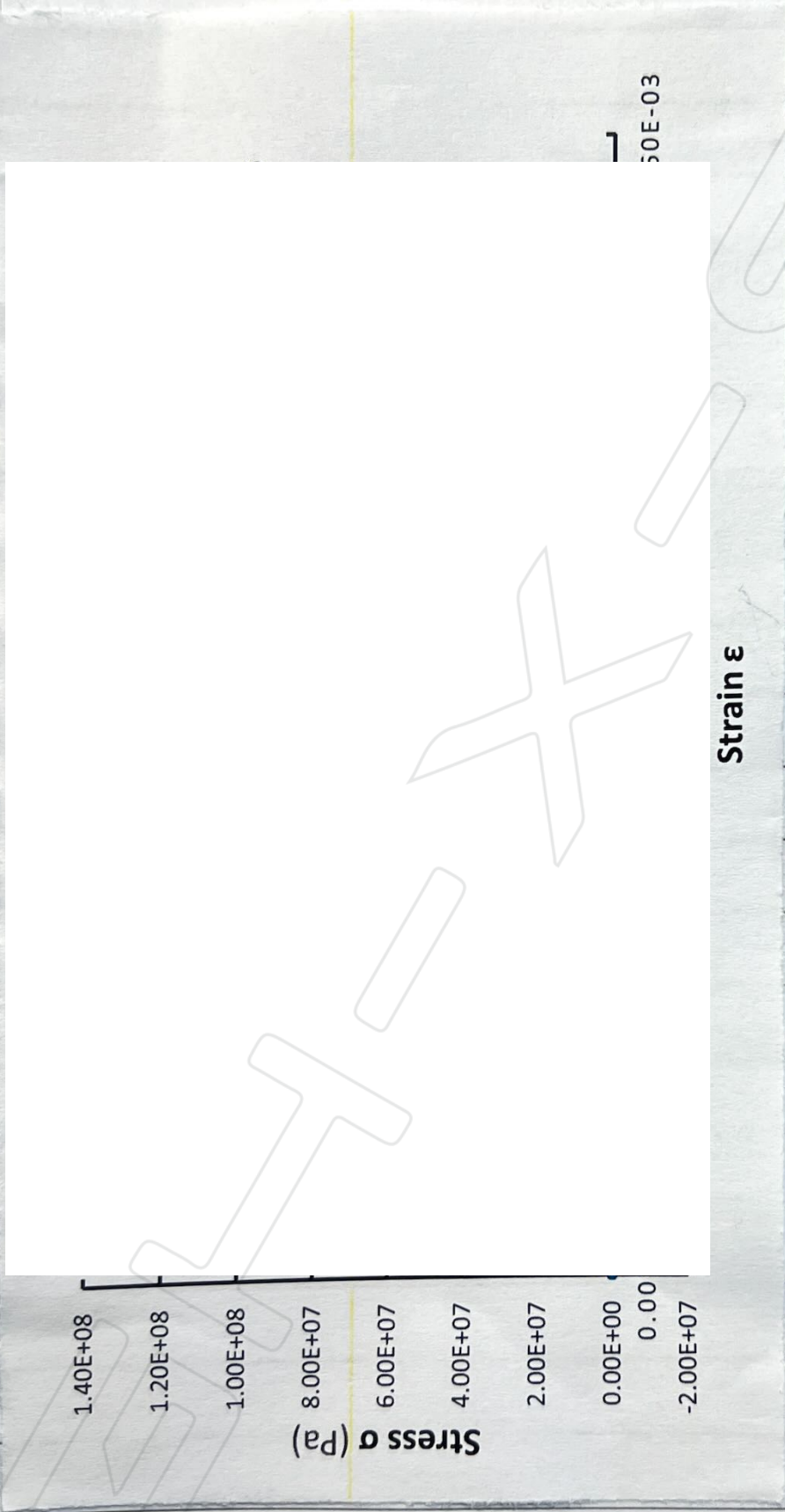
GAIN*Vg (V)

0.50
0.40
0.30
0.20
0.10
0.00
-0.10

Applied force F (kN)

4.00
0

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$$E_{Fe} > E_{Cu} > E_H > E_{Al}$$

$$E_{real} < E_{measured}$$